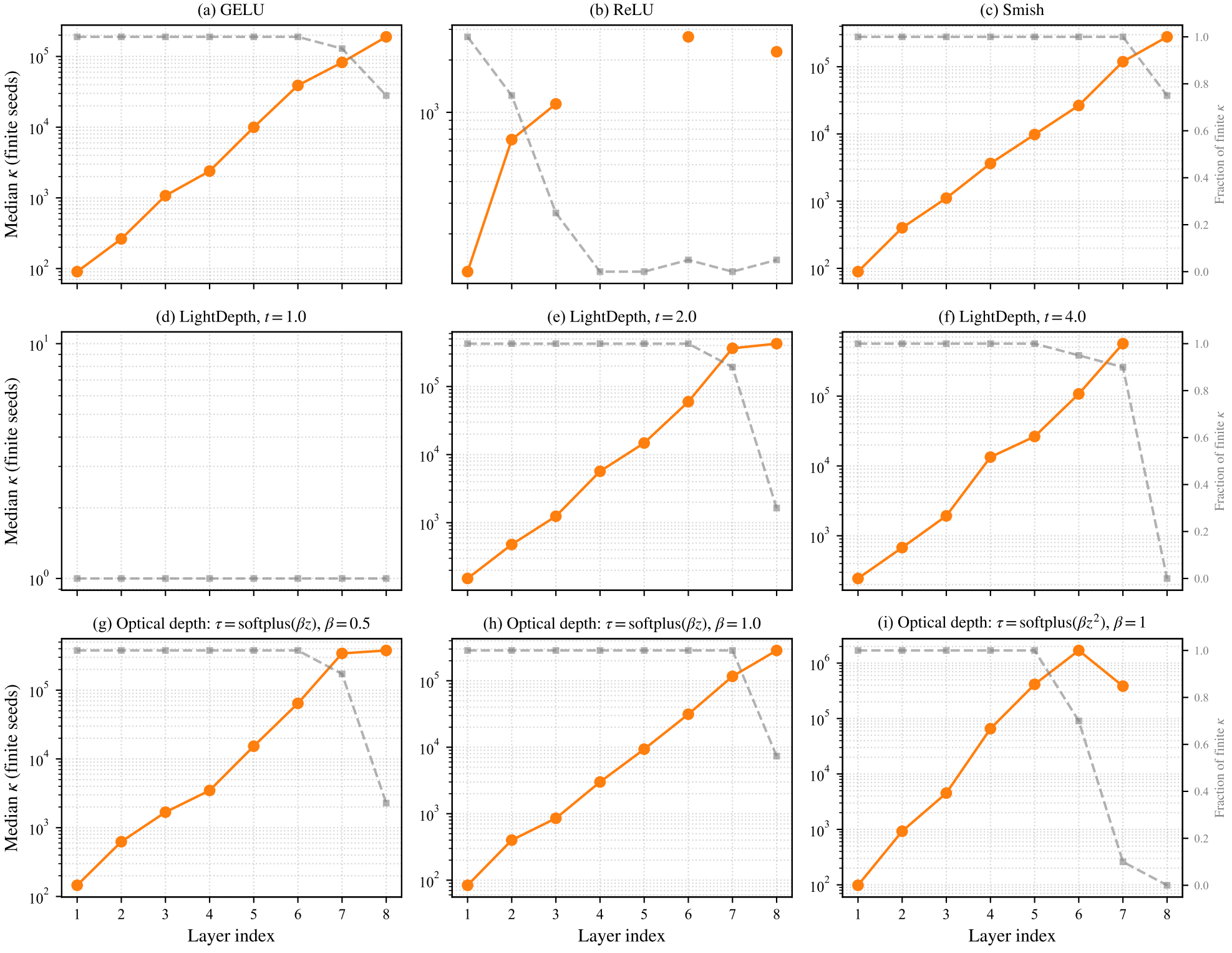
Condition number vs. depth ($N = 8$ layers; median κ over finite seeds)
(a) GELU
(b) ReLU
(c) Smish
(d) LightDepth, $t = 1.0$
(e) LightDepth, $t = 2.0$
(f) LightDepth, $t = 4.0$
(g) Optical depth: $\tau = \mathrm{softplus}(\beta z)$, $\beta = 0.5$
(h) Optical depth: $\tau = \mathrm{softplus}(\beta z)$, $\beta = 1.0$
(i) Optical depth: $\tau = \mathrm{softplus}(\beta z^2)$, $\beta = 1$
Median κ (finite seeds)
Fraction of finite κ
Layer index